

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and LangFlow”

Project Tile – [Fitness Buddy AI]

Domain of Project – [Healthcare]

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Problem Statement -

Problem Statement : Fitness Buddy AI

The Challenge

In today's fast-paced world, many individuals struggle to maintain a healthy lifestyle due to lack of personalized guidance, time constraints, and inconsistent motivation. Traditional fitness solutions often require expensive subscriptions, in-person consultations, or rigid schedules that don't adapt to personal preferences or daily routines.

There is a growing need for an accessible, friendly, and intelligent virtual assistant that can provide on-demand fitness advice, healthy lifestyle suggestions, and basic nutrition guidance—all tailored to individual needs and available at any time.

The Objective

The objective of Fitness Buddy AI is to provide users with personalized fitness guidance, workout recommendations, nutrition suggestions, and progress tracking through an intelligent AI-powered assistant.

The solution aims to:

- Generate personalized workout plans based on age, weight, fitness level, and goals.
- Provide real-time fitness coaching and motivation.
- Track daily activities, calories burned, and workout performance.
- Recommend healthy meal suggestions aligned with fitness goals.
- Monitor progress and adapt plans dynamically.
- Help users maintain consistency and achieve long-term fitness goals.

Technology Used

Tech Stack (Mandatory)

Use **any one** of the following approaches:

- LangFlow + IBM watsonx.ai + Granite Models
- IBM Cloud Agentic Lab with IBM Granite Models
- Replicate with LangChain + RAG + IBM Granite Models

I used LangFlow + IBM watsonx.ai + Granite Models

Proposed Solution -

Fitness Buddy AI is an Agentic AI-powered fitness assistant built using LangFlow, IBM watsonx.ai, and Granite Models. The system uses multiple intelligent agents working together to provide a complete fitness experience.

Key Features:

Personalized workout generation

- AI-based fitness coaching
- Nutrition recommendations
- Daily activity tracking
- Goal monitoring and progress analysis
- Motivational feedback and reminders
- Real-time fitness Q&A assistant

Core Agents:

1. Fitness Knowledge Agent

Retrieves workout information, exercise techniques, and fitness best practices using RAG.

2. Workout Planner Agent

Creates personalized workout routines based on user goals such as weight loss, muscle gain, endurance, or general fitness.

3. Nutrition Advisor Agent

Provides meal recommendations, calorie guidance, and hydration suggestions.

4. Progress Tracking Agent

Tracks completed workouts, calories burned, and fitness improvements.

5. Motivation & Wellness Agent

Provides reminders, encouragement, recovery tips, and healthy lifestyle recommendations.

Langflow component Used -

1.Chat Input

Collects user fitness goals, workout preferences, and health information.
Serves as the primary interaction point between the user and the AI system.

2. Prompt Template

Structures user input into optimized prompts for IBM Granite Models.
Ensures consistent and accurate AI responses.

3. IBM watsonx AI Agent

Processes user requests and coordinates multiple fitness-related tasks.
Acts as the central intelligence layer of the system.

4. IBM Granite-4.0-8B-Instruct Model

Generates workout plans, nutrition recommendations, fitness explanations, and motivational guidance.
Provides reasoning and personalization capabilities.

5. RAG Retriever

Retrieves fitness, exercise, and nutrition information from trusted knowledge sources.
Enhances accuracy and reliability of responses.

6. Vector Database (FAISS / ChromaDB)

Stores embeddings of fitness knowledge and workout resources.
Enables fast semantic search and information retrieval.

7. Memory Component

Maintains user context such as fitness goals, previous workouts, and progress history.
Supports personalized and continuous interactions.

Langflow workflow -

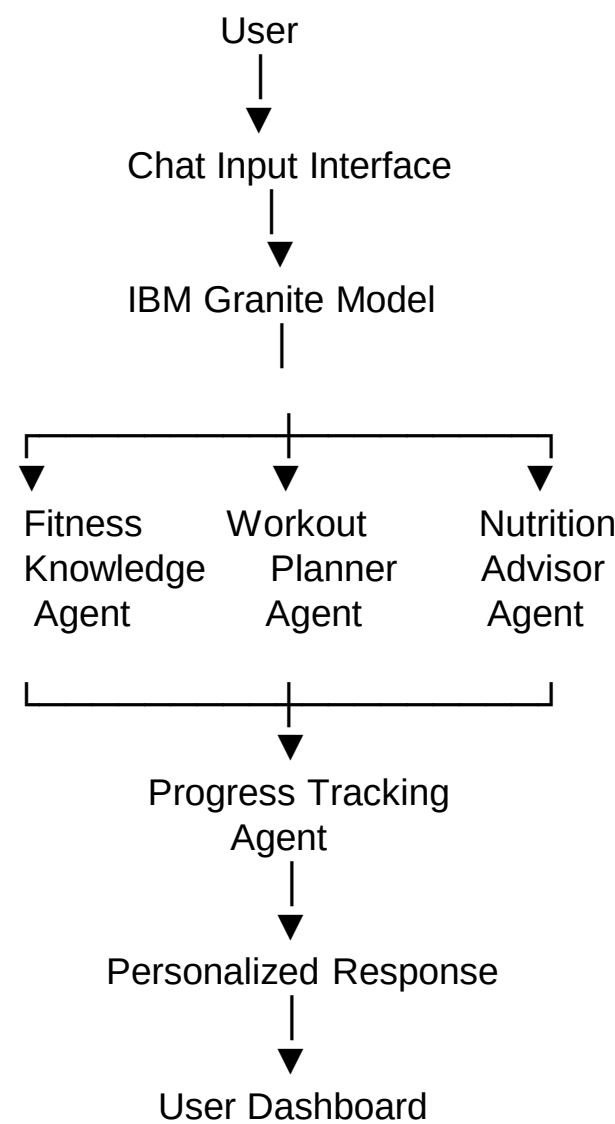
The screenshot displays the Langflow web application interface. At the top, the browser address bar shows the URL `localhost:7860/flow/96282b87-b61e-4798-ae06-b115809b1823`. The main header includes the Langflow logo, a 'Starter Project' dropdown, and a 'New Flow (1)' button. On the right side of the header, there are statistics for '149k' and '25k', along with a 'Playground' button and a 'Share' dropdown.

The left sidebar contains a search bar and a 'Components' panel. The 'Components' panel lists various categories: 'Input & Output', 'Data Sources', 'Models & Agents', 'LLM Operations', 'Files & Knowledge', 'Processing', 'Flow Control', 'Utilities', and a 'Discover more components' button. At the bottom of the sidebar is a '+ New Custom Component' button.

The central workspace shows a workflow diagram on a dark grid background. It starts with a 'Chat Input' component, which is connected to an 'IBM watsonx.ai' component. The 'IBM watsonx.ai' component has several input fields: 'Input' (labeled 'Receiving Input'), 'System Message' (labeled 'Type something...'), 'watsonx API Endpoint' (with the value `https://au-syd.ml.cloud.ibm.com`), 'watsonx Project_ID' (with the value `a3f564f0-15ea-4b20-b5f9-ad835...`), 'watsonx Space_ID' (labeled 'Type something...'), 'Watsonx API Key' (masked with dots), and 'Model Name' (with the value `ibm/granite-8b-code-instruct`). The output of the component is labeled 'Model Response'. This output is connected to a 'Chat Output' component.

At the bottom of the interface, there is a status bar showing the system temperature as '37°C Partly sunny', a search bar, and a taskbar with various application icons. The bottom right corner displays the date and time: '19:16 06-06-2026'.

Architecture blueprint



Role of Agentic AI in the solution

Agentic AI enables Fitness Buddy AI to act as an intelligent personal trainer rather than a simple chatbot.

How Agentic AI Helps

- Breaks complex fitness tasks into specialized agents.
- Enables collaboration between workout, nutrition, and progress-tracking agents.
- Understands user goals and context.
- Continuously adapts recommendations based on performance.
- Provides proactive suggestions and reminders.
- Delivers highly personalized fitness experiences.
- The multi-agent architecture allows Fitness Buddy AI to function as a virtual fitness coach available 24/7.



Technology Used

- **LangFlow**
Used to visually design and orchestrate multi-agent workflows.
- **IBM watsonx.ai**
Used for AI model deployment, prompt management, embeddings, and governance.
- **IBM Granite 4.0 Models**
Used for reasoning, fitness recommendations, summarization, and conversational interactions.
- **Retrieval-Augmented Generation (RAG)**
Used to retrieve accurate fitness and nutrition information from trusted sources.
- **Vector Database (FAISS / ChromaDB)**
Stores fitness knowledge and workout-related embeddings for fast retrieval.
- **IBM Cloud**
Provides scalable and secure infrastructure for deployment

Novelty and Uniqueness

Fitness Buddy AI differentiates itself from traditional fitness applications by combining **Agentic AI, Retrieval-Augmented**

- **Generation (RAG), and personalized fitness intelligence** into a single smart platform.
- **Multi-Agent Intelligence** – Multiple specialized AI agents collaborate to manage workouts, nutrition, progress tracking, and wellness guidance instead of relying on a single chatbot.
- **Hyper-Personalized Fitness Plans** – Generates customized workout and nutrition recommendations based on user goals, fitness level, age, body metrics, and lifestyle.
- **RAG-Based Knowledge Retrieval** – Retrieves accurate fitness and health information from trusted sources, reducing misinformation and improving recommendation quality.
- **Adaptive Learning System** – Continuously updates fitness plans based on user progress, workout history, and changing goals.
- **Real-Time Progress Monitoring** – Tracks performance, calories burned, workout completion, and fitness improvements with actionable feedback.
- **Motivational AI Coach** – Provides personalized encouragement, reminders, and habit-building strategies to improve consistency.

Git Hub Link

The complete project source code, documentation, architecture blueprint, and workflow are available in the GitHub repository.

with format – <https://github.com/rekha-1414/Fitness-Buddy-AI-Agentic.git>

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Future Scope

1. Wearable Device Integration

Connect with smartwatches and fitness bands to collect:

Heart rate

Step count

Sleep quality

Calories burned

This enables real-time adaptive workout plans.

2. Computer Vision Workout Analysis

Use camera-based posture detection to analyze exercise form and provide correction suggestions.

3. AI Voice Fitness Coach

Enable voice conversations for hands-free workout guidance.

4. Personalized Injury Prevention

Detect overtraining risks and recommend recovery plans.

5. Community & Social Fitness

Allow users to join fitness challenges and compete with friends.

6. Mental Wellness Integration

Combine fitness, mindfulness, stress management, and sleep coaching in one platform.

7. Predictive Health Analytics

Use AI to predict fitness outcomes and suggest preventive health actions.

8. Enterprise Wellness Programs

Thank You!

Thank you for your time and interest.